# Processors

## Hard Core!

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In only a matter of months, the processor industry changed its tune from clock speed to multi-cores—so quickly did this happen that it left tech-heads swooning. Intel was quick to drop speed from its DNA, since it was virtually humiliated over the humbling performance of its Netburst architecture against the mighty AMD K8.

The order of the day is dual-core processors—AMD's Athlon 64 X2 series with its integrated Southbridge and HyperTransport vs. Intel's Core Duo, and more importantly, its Core 2 Duo line. Intel is winning this particular performance battle with the Core 2 Duos. AMD has yet to refresh its aging K8 architecture, but is instead offering a redesign of its Opteron line in a quad-core configuration code-named "Barcelona," a few months from now. AMD has made it clear that we should expect the Barcelona processors to be 40 per cent faster than Intel's quad-core Xeons across "a wide variety of workloads." With Dell currently offering quad-core Opterons, AMD's offerings could become popular with the introduction of Barcelona. Indeed, AMD expects Barcelona to capture 30 per cent of the quad-core market, up from its current zero per cent share.

As of now, though, AMD has a dual-core, dual-socket option for the enthusiast market with a tweakable BIOS and SLI support, called the Quad FX platform (a.k.a. 4x4). AMD should also drop its processor prices across the board by the time you read this. Mostly, the drop will be to make way for the Athlon 64 X2 6000+ processor, which is expected any time soon as of this writing. Price drops are rumoured to be significant across the board: the Athlon 64 X2 3800+ is said to drop from $138 to $113, the Athlon 64 X2 4600+ from $215 to $195, and the Athlon 64 X2 5200+ to just $222 from its current price of $295. Note that these are suggested distributor prices. With these cuts, AMD will still remain the best processor you can buy, offering the best value in terms of both price and performance. Factor in pure speed, though, and Intel is still champion.

Quad-core processors from Intel already offer a significant advantage over their dual-core siblings, and we would generally recommend that you go in for a slightly slower quad-core solution than a faster dual-core one, especially if you are a gamer or a graphics/3D designer. Intel is also looking to introduce faster FSB to its processor range, this will tie in closely with the introduction of new chipsets, of course: Intel will introduce Core 2 models ending in "50" (Core 2 Duo E6850 and E6650) which will take advantage of the 1333 MHz FSB. The newer cores will only officially be supported on Intel's upcoming x35 series of chipsets. Meanwhile, the low-end 6000 series will also receive a cache boost—a 4 MB L2 cache will give a nice performance boost like it does their more expensive brethren. Prices will remain the same.

Meanwhile, the K10 core from AMD should burrow out of its secret underground bunker—competition is good, and we can't imagine AMD letting Intel get away with the performance crown. Intel, though, is already drumming up plans for 80-core processors against the silence of its arch-rival: these multi-core units are not expected for another five years, but it is heartening to see their game plan so early in the cycle.

One of the biggest problems for multi-core in the future would be that of memory bandwidth. Interestingly, graphics cards faced much the same problem and solved them by throwing extremely fast and expensive memory chips at the GPU, along with a wide and speedy bus—scaling such a solution to say 16, 32 or more cores for a CPU becomes prohibitively expensive and impractical.

AMD will likely base a solution to this problem around its HyperTransport bus, while Intel has revealed its future direction through its "Terascale" project: instead of transistors arranged flat on a die, this chip's design consists of 80 tiles laid out in an 8 x 10-block array. Each tile carries a small core that can run a simple instruction set for processing floating-point data. The tile also includes a router connecting the core to an on-chip network that links all the cores to each other and gives them access to memory.

What is tremendously exciting is to imagine a multi-core processor integrate itself with a graphics processor. Both AMD with its Fusion project and Intel with Larrabee are headed in that direction—a massively multi-core processor connected to a large amount of on-die memory and surrounded by specialist vector units—probably the ultimate CGPU!